OPPORTUNITY BRIEF · SALES AND PRESALES

Edge-based anomaly detection for rail network signaling to prevent disruptions from cyber and physical events.

Transportation and Logistics × Critical infrastructure monitoring × Edge computing · OS074 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
49 is the world moving	**46** can we play, can we win	**MEDIUM** 1 named in the evidence	**€824m** partial evidence · per year	**NEXT** pilots published no volume pr...	**L1** 4 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying edge-based anomaly detection for rail signaling to prevent disruptions from both cyber and physical events. It targets rail operators and infrastructure managers who must protect critical transport networks. The need is urgent because transport networks are critical infrastructures essential to the Single Market, and recent publications highlight the growing focus on resilience against cyber and non-cyber events.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€1.3bn** €290m – €7.7bn	**€824m** €180m – €4.8bn	**€9.9m** €2.2m – €57.7m	partial
Observed: tendered contract value, annualised	€115m	€115m	€1.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Edge computing	11.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

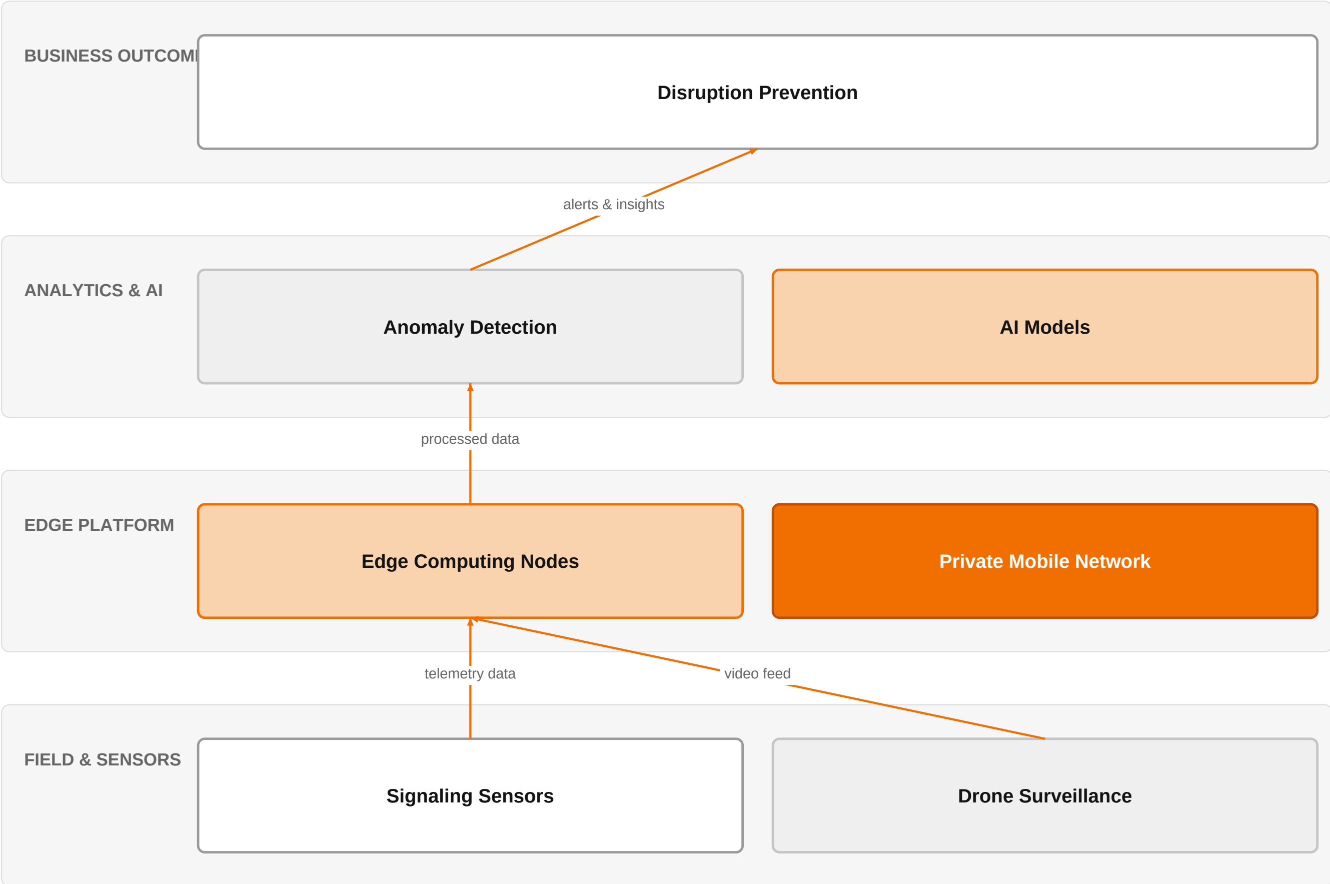
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Edge Anomaly Detection for Rail Signaling



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how edge-based anomaly detection can provide real-time insights to prevent disruptions in rail signaling.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The transport network is recognized as a Critical Infrastructure, essential to the Single Market, and is a large-scale interconnected system. Traditional cloud-based solutions for IoT fault detection suffer from latency and energy overhead, driving interest in edge-based AI. Recent publications from Microsoft and Rail Engineer indicate a shift toward AI-driven operating models and increased attention to cybersecurity in rail. These changes create a window for edge-based anomaly detection solutions.

`Sources: SIG-B0A75430817E cordis.europa.eu 2024-09-01; SIG-F48074B55B20 openalex.org 2025-10-16; SIG-C765B2D09832 Microsoft 2026-07-16; SIG-E66DFDD2061F Rail Engineer 2026-04-27`

Who buys, and why now

The buyer is typically the Chief Technology Officer or Head of Infrastructure at a rail operator or infrastructure manager. The pain is felt by operations and security teams who must ensure continuous service and protect against disruptions. The trigger is often a recent cyber incident, a regulatory audit, or a near-miss physical event that exposes vulnerabilities in legacy signaling systems. They need to detect anomalies early to prevent costly delays and safety risks.

Why it is hot now — every claim bound to a dated source

- The transport network is among the so-called Critical Infrastructures, essential to maintaining vital functions of the Single Market, and is a large-scale interconnected and interdependent system. [SIG-B0A75430817E]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Cybersecurity experts, IoT experts, and Digital experts to design and deploy edge-based anomaly detection. The engagement would start with a security assessment of the signaling network, followed by deploying edge computing nodes (potentially using Private mobile networks for connectivity) and integrating AI models for anomaly detection. Orange Drone Guardian could be offered for physical perimeter monitoring. The solution would be delivered in phases, starting with a pilot on a specific line, then scaling.

Why Orange can win

Orange can win because it combines deep cybersecurity expertise with IoT and digital capabilities, and has experience in critical infrastructure protection. The ability to provide secure, sovereign solutions is a differentiator. However, the assets are thin for this specific edge computing use case; Orange does not have a named edge AI platform, so it would need to leverage partners like NVIDIA or HPE for hardware and AI acceleration.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L1	offer addresses the use case but not this technology	unconfirmed
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Microsoft, which is already promoting an 'AI Railroad Brain' concept, and AWS, which sells edge computing. Deutsche Telekom/T-Systems and Lumen Technologies compete in edge computing and cybersecurity. NTT DATA offers edge computing and connectivity solutions. Schneider Electric specializes in industrial OT edge computing. HPE and Nokia also sell edge computing and are active in transportation. Each has strengths: Microsoft in AI and cloud, AWS in scalability, telecom peers in connectivity, and industrial players in OT integration.

`Sources: SIG-C765B2D09832 Microsoft 2026-07-16`

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-C765B2D09832
AWS	Hyperscaler	sells Edge computing — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions, Cybersecurity	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions, Cybersecurity	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Connectivity Solutions	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions — also an Orange partner	structural	—

Nokia	Network equipment vendor	sells Edge computing; active in Transportation and Logistics; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. Have you experienced any recent cyber or physical incidents that disrupted your signaling operations?
2. What is your current approach to monitoring signaling infrastructure for anomalies?
3. Do you have any existing edge computing infrastructure deployed in your network?
4. What are your key performance indicators for signaling uptime and disruption response?
5. How do you currently handle data from IoT sensors in your signaling system?
6. Are you subject to any specific regulations or standards for critical infrastructure protection?

Objections you will meet

They will say	You can answer
We already have a security operations center and monitoring tools.	That's true, but traditional cloud-based monitoring can introduce latency and energy overhead. Edge-based anomaly detection can provide real-time insights at the source, reducing response times and improving reliability.
Edge AI is too complex and risky for safety-critical rail systems.	We understand the safety concerns. That's why we propose a phased approach, starting with a pilot on a non-critical line, and we work with partners like NVIDIA and HPE to ensure robust, tested solutions.
We are already working with a hyperscaler on digital transformation.	That's fine. We can complement existing efforts by providing a sovereign, secure edge layer that integrates with your current cloud strategy, leveraging our cybersecurity expertise.

Risks and unknowns

The main risk is that rail operators may be cautious about adopting edge AI due to safety certification requirements and long procurement cycles. There is also uncertainty about the performance of edge AI in real-world rail environments, as the evidence is based on pilots and research. Additionally, the competitive landscape is crowded, and Orange's edge computing capabilities are not yet proven in this vertical.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a rail operator's critical infrastructure monitoring tender, differentiating Orange by integrating Orange Drone Guardian for physical perimeter monitoring and private mobile networks for secure edge connectivity, and citing Point One Navigation as a reference for precise positioning.
Sales	In a conversation with a rail network operator's head of operations, say: 'We can help you detect anomalies in your signaling network at the edge, combining our cybersecurity expertise with private mobile networks to prevent disruptions from both cyber and physical events.'
Strategist / Innovator	Commission a deep-dive brief on edge-based anomaly detection for rail signaling, leveraging Orange Group researchers and cybersecurity experts to map the threat landscape and pilot evidence, and assess whether to invest in a prototype with NVIDIA, Dell, or HPE as edge partners.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-B0A75430817E	2024-09-01	cordis.europa.eu	1	Transport resilience against Cyber and Non-Cyber Events to prevent Network Disruption
SIG-C765B2D09832	2026-07-16	Microsoft	2	The AI Railroad Brain: A new operating model for freight rail - Microsoft
SIG-E66DFDD2061F	2026-04-27	Rail Engineer	2	Cyber security update - Rail Engineer

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:40+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.